

Aerospace Academy Version

Unit 5

Materials, Structures & Prototype Testing

Unit Overview

Students investigate how material choice, geometry, load paths, and testing affect lightweight aerospace structures. Students build and test prototypes, collect data, analyze failure, and propose redesigns.

Essential Question

How do engineers design lightweight structures that are strong enough to survive real loads?

Name: _____ Period: _____ Date: _____

Learning Targets

1. Explain the major concepts of materials, structures & prototype testing.
2. Apply engineering documentation, sketching, modeling, testing, or communication skills to an aerospace-focused task.
3. Use evidence and reflection to improve design work.
4. Submit a complete unit portfolio or project package.

Unit Deliverables

- Materials comparison
- Structure sketch
- Test plan
- Prototype
- Test data table
- Failure analysis
- Redesign proposal
- Structures report

Vocabulary

Term	Definition
Material	
Structure	
Load	
Load Path	
Compression	
Tension	
Shear	
Torsion	
Strength-to-Weight Ratio	
Prototype	
Test Plan	
Data	
Failure Mode	
Redesign	
Factor of Safety	

Unit Project: Lightweight Aerospace Structure Test

You will apply the skills from this unit to create an aerospace-focused project package. Your work should show clear documentation, professional effort, evidence-based decisions, and reflection.

Day 1 — Aerospace Structures

Objective: I can explain why aerospace structures must be lightweight and strong.

Bellwork: Why is weight such a big issue in aerospace design?

Core Task: Analyze examples of trusses, ribs, spars, frames, and brackets.

Evidence of Work / Notes:

Exit Ticket: What structure feature saves weight?

Day 2 — Materials and Trade-Offs

Objective: I can compare materials using engineering criteria.

Bellwork: What makes one material better than another?

Core Task: Compare material options by strength, weight, cost, availability, and manufacturability.

Evidence of Work / Notes:

Exit Ticket: What material would you choose first?

Day 3 — Loads and Load Paths

Objective: I can identify how forces move through structures.

Bellwork: Where does force go when a structure is loaded?

Core Task: Sketch load paths through simple structures.

Evidence of Work / Notes:

Exit Ticket: Where is failure most likely?

Day 4 — Strength-to-Weight Ratio

Objective: I can explain strength-to-weight ratio.

Bellwork: Is the strongest design always best?

Core Task: Calculate or compare performance per unit mass.

Evidence of Work / Notes:

Exit Ticket: Why is ratio useful?

Day 5 — Prototype Testing

Objective: I can explain why engineers test prototypes.

Bellwork: What can testing show that sketches cannot?

Core Task: Review testing expectations and safety rules.

Evidence of Work / Notes:

Exit Ticket: What should a test measure?

Day 6 — Test Plan Development

Objective: I can create a test plan for a structure.

Bellwork: Why should engineers plan tests before building?

Core Task: Write a test objective, variables, procedure, and data table.

Evidence of Work / Notes:

Exit Ticket: What is your test objective?

Day 7 — Structure Concept Sketches

Objective: I can sketch structural concepts before building.

Bellwork: How can geometry make a structure stronger?

Core Task: Create at least three structural concepts and label load paths.

Evidence of Work / Notes:

Exit Ticket: Which geometry seems strongest?

Day 8 — Prototype Build Day 1

Objective: I can build a prototype safely and document decisions.

Bellwork: What must your team document while building?

Core Task: Begin building the structure using approved materials.

Evidence of Work / Notes:

Exit Ticket: What design choice did you make?

Day 9 — Prototype Build Day 2

Objective: I can revise a prototype during construction.

Bellwork: What should trigger a design change?

Core Task: Continue building and record revisions.

Evidence of Work / Notes:

Exit Ticket: What changed and why?

Day 10 — Testing Day

Objective: I can collect test data safely and accurately.

Bellwork: What makes test data trustworthy?

Core Task: Test structures, record load, mass, failure point, and observations.

Evidence of Work / Notes:

Exit Ticket: What happened during failure?

Day 11 — Failure Analysis

Objective: I can analyze why a structure failed.

Bellwork: Is failure bad if it gives information?

Core Task: Identify failure mode, weak points, and design causes.

Evidence of Work / Notes:

Exit Ticket: What was the main failure mode?

Day 12 — Redesign Planning

Objective: I can propose improvements based on data.

Bellwork: How should data influence redesign?

Core Task: Create a redesign sketch and justify changes.

Evidence of Work / Notes:

Exit Ticket: What will your redesign improve?

Day 13 — Retest or Simulation Discussion

Objective: I can compare original and redesigned structures.

Bellwork: How would you prove the redesign is better?

Core Task: Plan a retest or explain expected performance improvement.

Evidence of Work / Notes:

Exit Ticket: What evidence would support your redesign?

Day 14 — Structures Report Work Day

Objective: I can communicate test results in a report.

Bellwork: What should a test report include?

Core Task: Assemble data, photos/sketches, failure analysis, and redesign proposal.

Evidence of Work / Notes:

Exit Ticket: What section needs more detail?

Day 15 — Structures Report Submission

Objective: I can submit and reflect on a structures testing report.

Bellwork: What did testing teach you about design?

Core Task: Submit report and complete reflection.

Evidence of Work / Notes:

Exit Ticket: What would you change in the next prototype?

Unit 5 Submission Checklist

Item	Complete?
Materials comparison	☐
Structure sketch	☐
Test plan	☐
Prototype	☐
Test data table	☐
Failure analysis	☐
Redesign proposal	☐
Structures report	☐

Unit 5 Project Rubric

Category	Points
Technical accuracy and completeness	20
Engineering process and documentation	20
Aerospace connection and design intent	15
Quality of sketches, models, data, or drawings	20
Use of evidence, testing, or decision making	15
Reflection and professionalism	10
Total	100

Unit 5 Student Work Templates

Structure Concept Sketch

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name: _____

Scale: _____

Purpose: _____

Date: _____



Notes / annotations: _____

Prototype Testing Data Table

Purpose: Testing and analysis

Directions: Record measurements clearly. Include units, observations, failure points, and redesign decisions based on evidence.

Trial	Measurement / Result	Observation	Change Needed
1			
2			
3			
4			
5			

Analysis / redesign notes:

Failure Analysis Template

Purpose: Learning from testing

Directions: Describe what failed, where it failed, why it may have failed, and what evidence supports your explanation.

Failure location _____

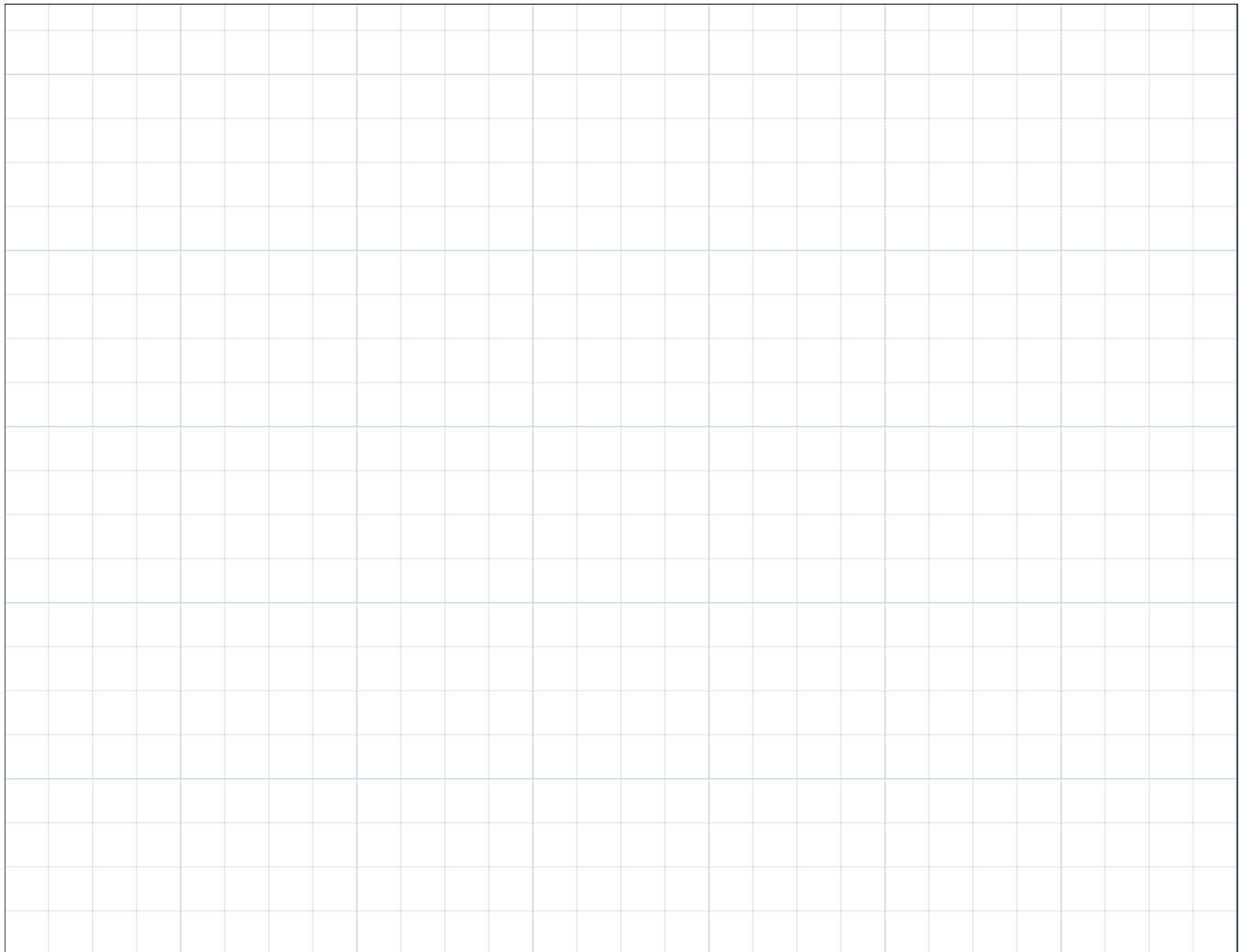
Failure type _____

Evidence observed _____

Likely cause _____

Redesign idea _____

Detailed notes / labeled failure sketch:



Redesign Sketch

Purpose: Isometric sketching

Directions: Use the isometric grid to show height, width, and depth. Keep edges aligned to the isometric axes unless intentionally sketching a curve. Add labels and annotations.

Part name: _____

Scale: _____

Purpose: _____

Date: _____



Notes / annotations: _____